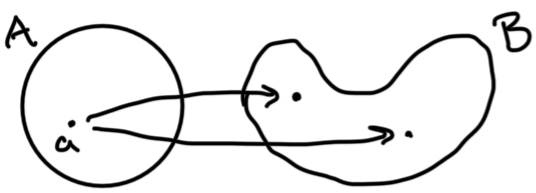
① Homework 1 (and Honors Assignment 1) are posted, due Friday (Jan 30) at 1:20pm

② Add the Honors Option by Friday, if you are interested

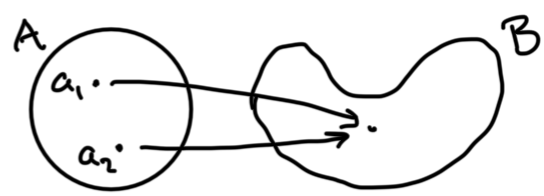
③ I highly recommend that you type your homework using LaTeX – I put some resources on Canvas to help you get started

## FUNCTIONS

<u>Def</u> Let $A, B$ be nonempty sets. A <u>function</u> $f: A \to B$ is a way of associating to each element $a \in A$ exactly one element in $B$, denoted $f(a)$.




Not a function



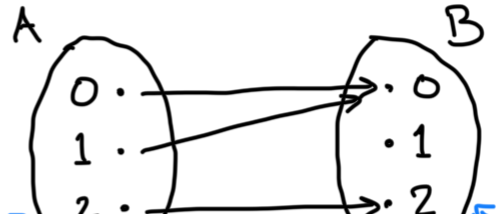

OK!

$A$ is called the <u>domain</u> of $f$.
$B$ is called the <u>range</u> of $f$.

<u>Ex</u> Let $A = \{0, 1, 2\} = B$. Are the following functions from $A$ to $B$?

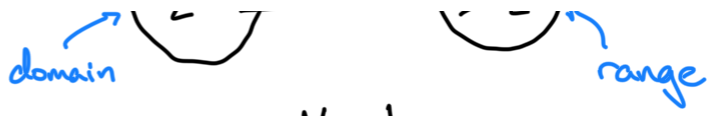
① $f(x) = x^2 - x$.

$f: A \to B$
$0 \mapsto 0$
$1 \mapsto 0$



$1 \mapsto 0$

$2 \mapsto 2$



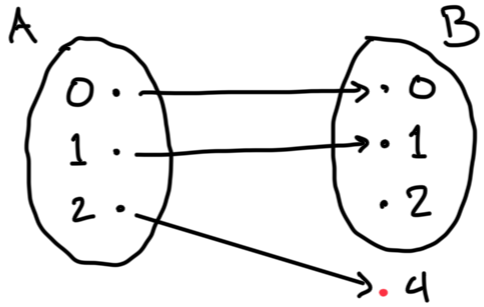

Yes!

② $f(x) = x^2$

$0 \mapsto 0$

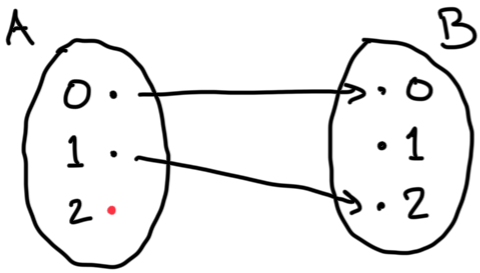
$1 \mapsto 1$

$2 \mapsto 4$



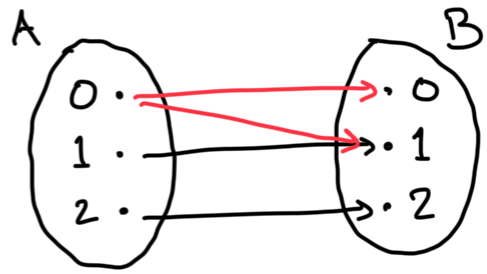

No.

③



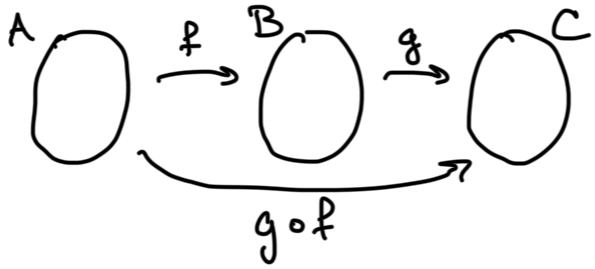

No.

④




No.

<u>Def</u> Let $A, B, C$ be nonempty sets and let $f: A \to B$, $g: B \to C$. The <u>composition</u> of $g$ with $f$ is the function $g \circ f : A \to C$, $(g \circ f)(a) = g(f(a))$.




<u>Ex</u> Let $f, g : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x$, $g(x) = x+1$. Then

$$(g \circ f)(x) = g(f(x)) = g(2x) = 2x+1$$
$$(f \circ g)(x) = f(g(x)) = f(x+1) = 2(x+1)$$

Not equal!

<u>Rmk</u> Composition of functions is not commutative. However, it is associative: $(h \circ g) \circ f = h \circ (g \circ f)$.

**Def** Let $f: A \to B$ be a function.

① If $X \subseteq A$, the <u>image of $X$</u> under $f$ is

$$f(X) = \{f(a) : a \in X\}.$$

② The <u>image</u> of $f$ is $f(A)$.

③ If $Y \subseteq B$, the <u>preimage of $Y$</u> under $f$ is

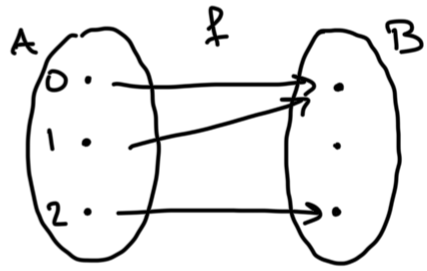
$$f^{-1}(Y) = \{a \in A : f(a) \in Y\}.$$

**Rmk** Here, "$f^{-1}$" is not the same thing as the inverse function of $f$!

- $f^{-1}(Y)$ is well-defined, even when there does not exist an inverse function $f^{-1}: B \to A$.
- $f$ and $f^{-1}$ do not cancel in general.

**Ex** $A = \{0, 1, 2\} = B$,

$$f: A \to B$$
$$0 \mapsto 0$$
$$1 \mapsto 0$$
$$2 \mapsto 2$$




$f(\{1, 2\}) = \{0, 2\}$

$f^{-1}(\{1, 2\}) = \{2\}$

$f(f^{-1}(\{1, 2\})) = f(\{2\}) = \{2\} \neq \{1, 2\}$

$f^{-1}(f(\{1, 2\})) = f^{-1}(\{0, 2\}) = \{0, 1, 2\} \neq \{1, 2\}$

**Exc** Let $f: A \to B$ be a function, and let $A_1, A_2 \subseteq A$ and $B_1, B_2 \subseteq B$. Then:

① $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$

② $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$

③ $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$

④ $f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$

⑤ $f^{-1}(f(A_1)) \supseteq A_1$

⑥ $f(f^{-1}(B_1)) \subseteq B_1$

<u>Pf of ⑥</u>: Pick any $y \in f(f^{-1}(B_1))$. Want: $y \in B_1$.
As $y \in f(f^{-1}(B_1)) = \{f(a) : a \in f^{-1}(B_1)\}$, $\underline{\exists}\, x \in f^{-1}(B_1)$
"there exists"
<u>s.t.</u> $f(x) = y$. As $x \in f^{-1}(B_1) = \{a \in A : f(a) \in B_1\}$,
"such that"
then $f(x) \in B_1$. Altogether, we conclude $y = f(x) \in B_1$,
as desired. $\square$

<u>Def</u> Let $f : A \to B$ be a function.

① $f$ is <u>surjective</u> (or <u>onto</u>) if $f(A) = B$.
I.e., $\forall b \in B\ \exists a \in A$ s.t. $f(a) = b$.

② $f$ is <u>injective</u> (or <u>one-to-one</u>) if it satisfies:
if $a_1, a_2 \in A$ s.t. $f(a_1) = f(a_2)$, then $a_1 = a_2$.

③ $f$ is <u>bijective</u> if it is injective and surjective.

<u>Rmk</u> These properties depend not only on $f$, but also on the domain $A$ and range $B$.

<u>Ex</u> $f(x) = 2x$

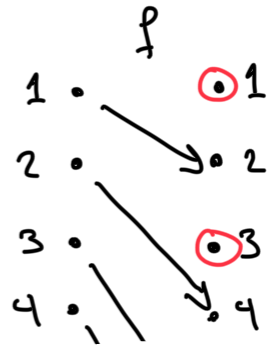
① $f : \mathbb{N} \to \mathbb{N}$ is...

- injective: For $m, n \in \mathbb{N}$,

  $f(m) = f(n) \Rightarrow 2m = 2n \Rightarrow m = n$

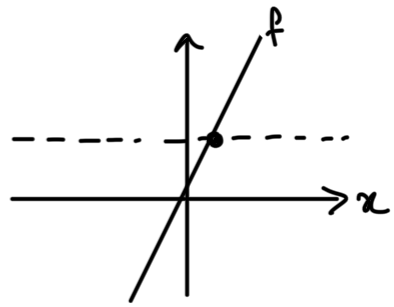
- not surjective: $1 \in \mathbb{N}$, but

  $\nexists\, n \in \mathbb{N}$ s.t. $f(n) = 1$




② $f : \mathbb{R} \to \mathbb{R}$ is...

- injective: Same proof works for $m, n \in \mathbb{R}$

- surjective: Fix $y \in \mathbb{R}$. Then $\frac{y}{2} \in \mathbb{R}$ satisfies $f(\frac{y}{2}) = y$.

- bijective